Improving Misaligned Multi-modality Image Fusion with One-stage Progressive Dense Registration

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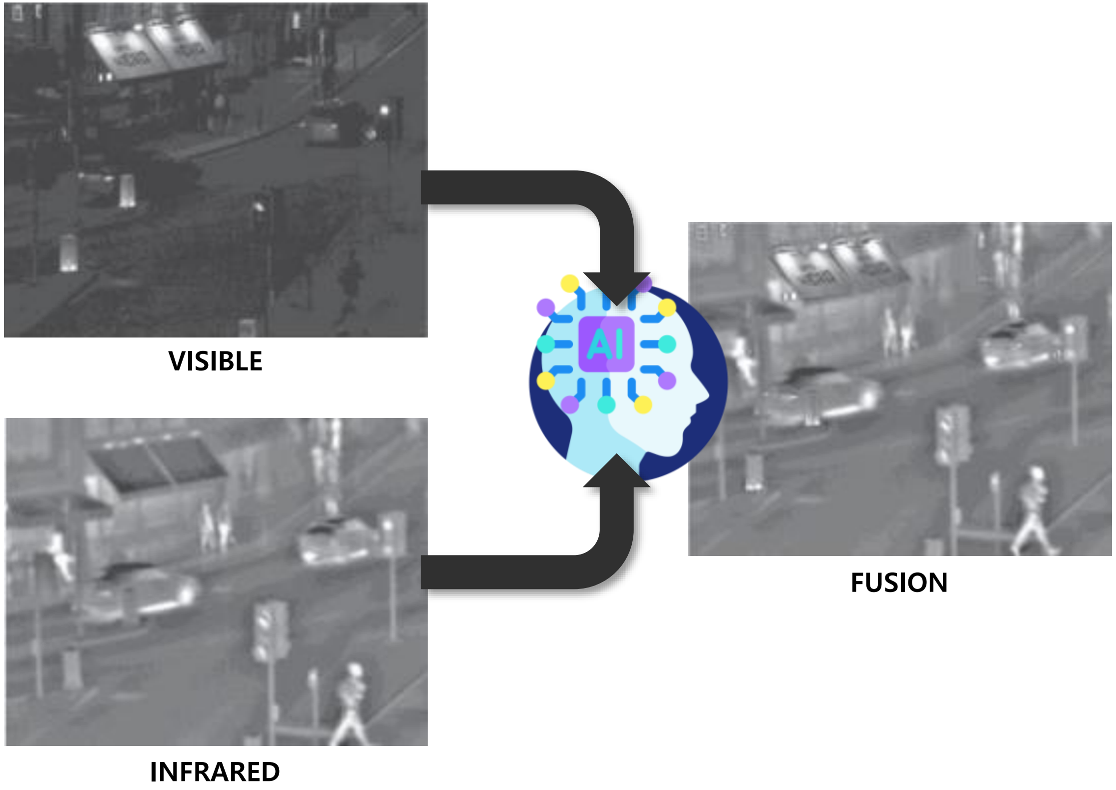
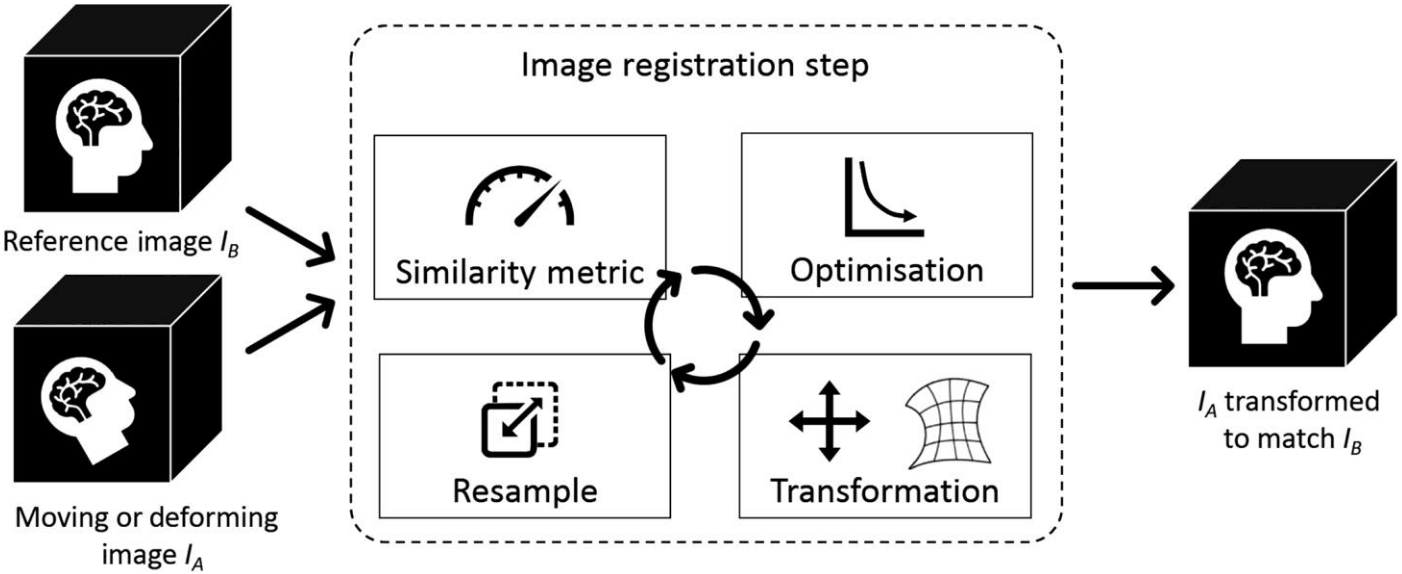


01. Introduction

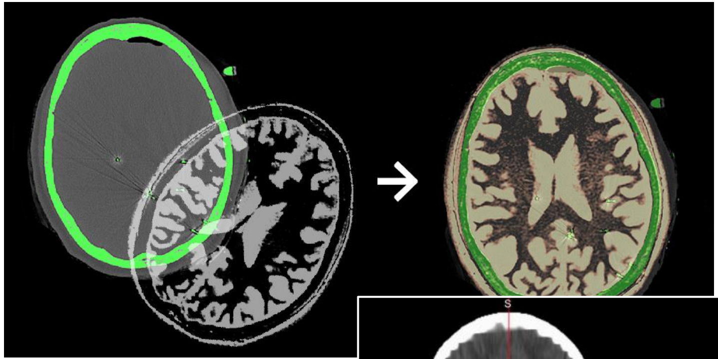
2. Image Registration

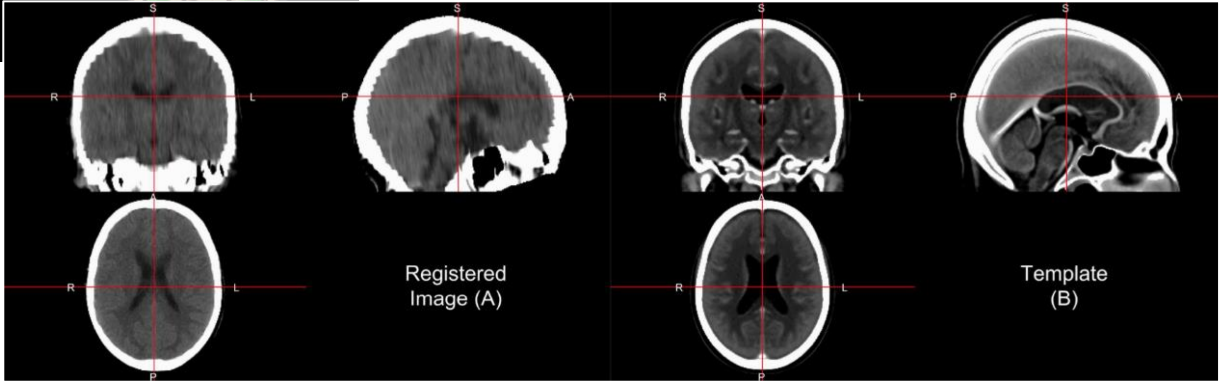


출처 : https://pebpung.github.io/registration/2022/01/30/Registration-1.html

01. Introduction

2. Image Registration





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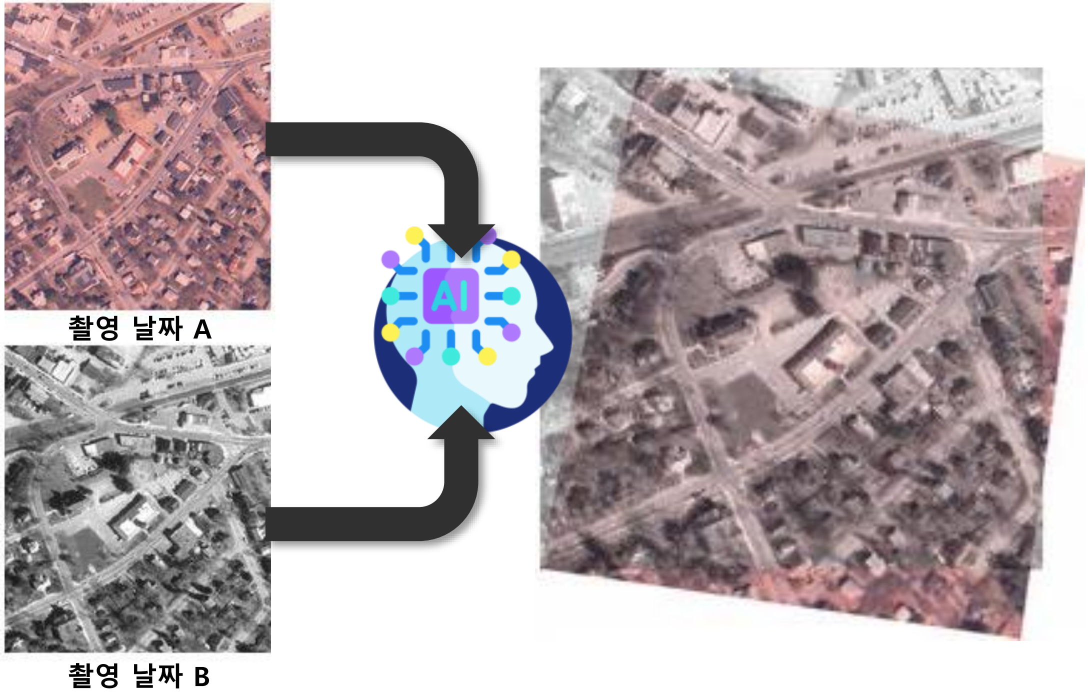
2. Image Registration



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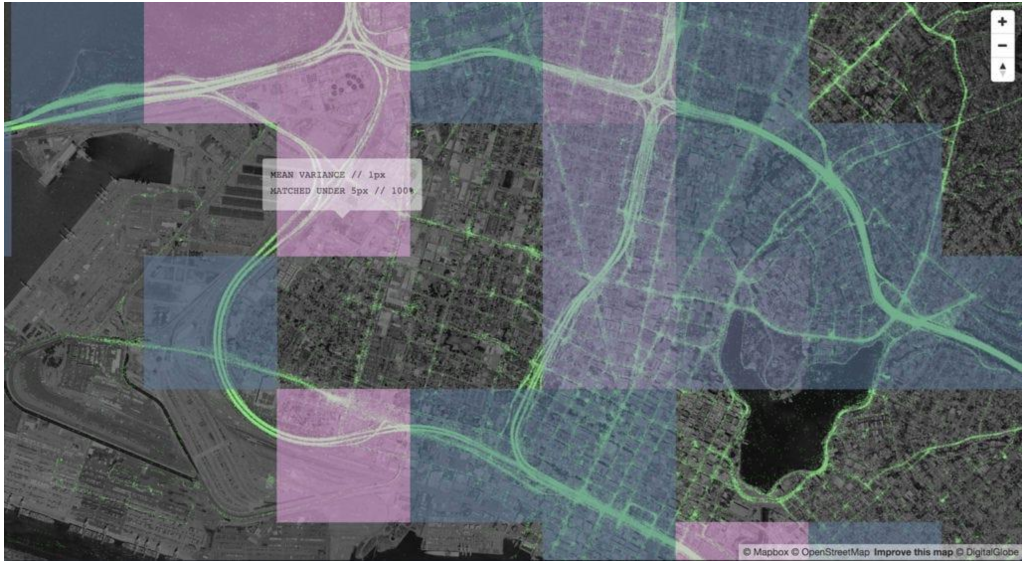


위성영상 업데이트 시 기존 객체와의 틀어짐

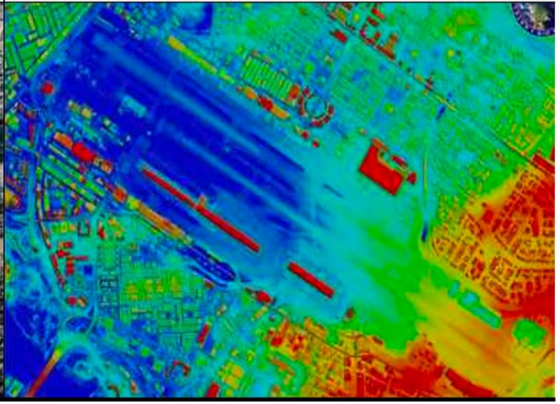
01. Introduction

2. Image Registration

Google Timelapse (부산)

01. Introduction

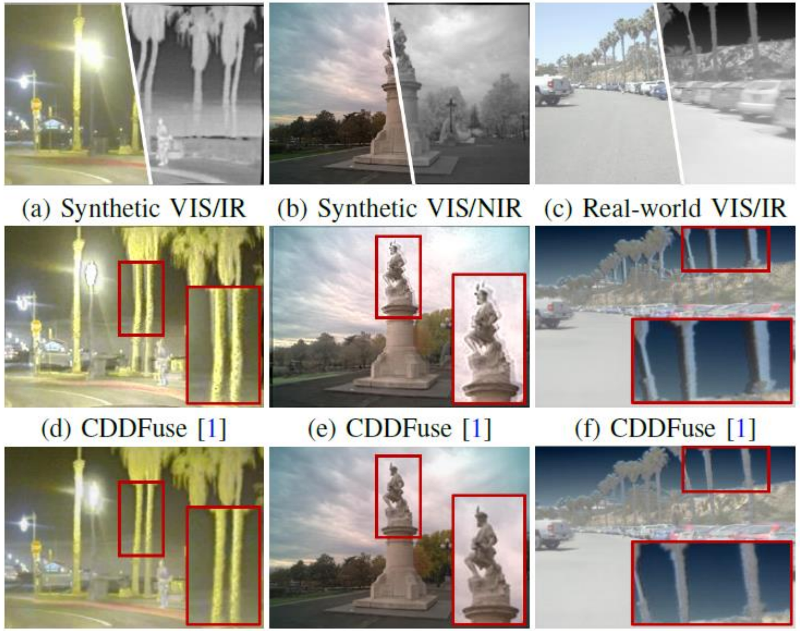
2. Image Registration

영상레이더(SAR)	전자광학(EO)	적외선(IR)
미국 앨버키키 국제공항		일반 공항
		

Remote Sensing에서 사용되는 대표적 센서 종류

01. Introduction

3. Proposed Method



(a) Synthetic VIS/IR (b) Synthetic VIS/NIR (c) Real-world VIS/IR

(d) CDDFuse [1] (e) CDDFuse [1] (f) CDDFuse [1]

(g) IMF (Ours) (h) IMF (Ours) (i) IMF (Ours)

1 Image Misalign으로 발생하는 Image Fusion 문제 해결

- Edge Ghost
- Structure Distortion

2 Coarse-to-Fine 2단계 Registration을 1단계로 제시

3 Local/Global Feature를 동시 반영하는 Image Fusion Network

02

Previous Work

02. Previous Work

1. Image Registration

- **Mono-modality Image Registration**
  - Optical Flow 기반 방법이 많이 쓰이나, Multi-modality Registration에서는 부적합
  - FlowNet (Ilg et al., 2017), VoxelMorph (Balakrishnan et al., 2018)
  - DGC-Net (Melekhov et al., 2019), GLU-Net (Truong et al., 2020)
- **Cross-modality Image Registration**
  - Optical Flow를 다중 Modality에 적용하려는 시도
  - Nemar (Arar et al., 2020), CrossRAFT (Zhou et al., 2022)

Optical Flow 예시

2. Image Fusion

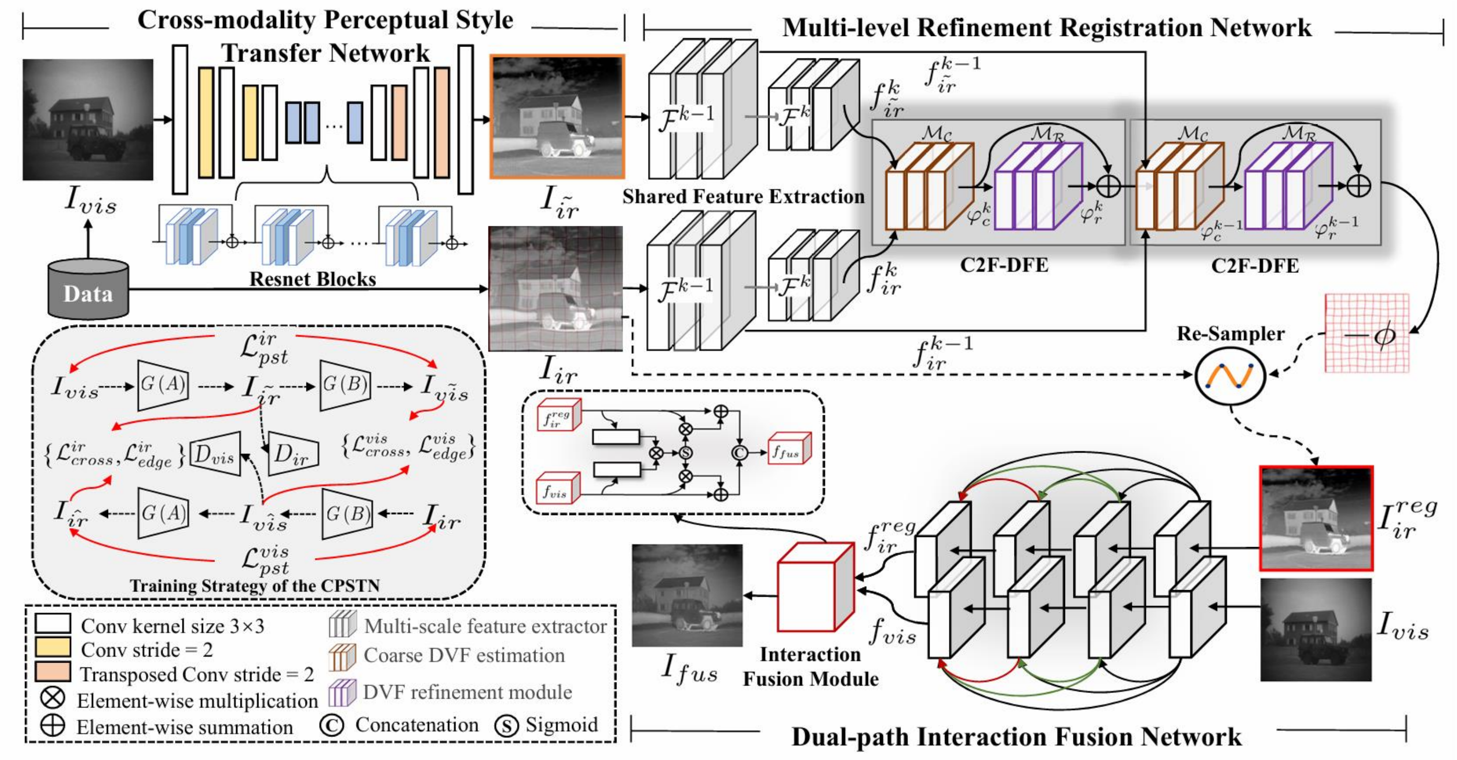
- **Deep Learning 기반 Image Fusion**

Method	Paper	Properties
DCN Densely Connected Network	DIDFuse,U2Fusion, DenseFuse, FusionDN …	• 두 Modality의 Feature를 효과적으로 추출하기 위해 Dense connection 구조를 활용하여 다양한 해상도의 feature를 결합하나 복잡하고 미세한 구조적 정보를 충분히 보존하지 못해 Fusion 이미지의 디테일 하락
GAN Generative Adversarial Network	FusionGAN, DDcGAN	• GAN 기반 기법을 통해 시각적으로 뛰어난 융합 이미지 생성 • 한계: Handcrafted design 요소가 많아 Redundancy가 발생, 네트워크의 유연성이 떨어져 다양한 상황에 대해 강건하게 대응 불가
NAS Nueral Architecture Search	SMOA	• 네트워크 구조를 탐색하여 효과적인 융합 방식을 학습
Transformer	SwinFusion, CGTF, PT Fusion	• Transformer를 사용하여 Long-range Feature를 이용하고 융합 이미지 품질 개선
Seamantic-aware Image Fusion	TarDAL	• 객체 검출, 장면 이해 등의 downstream task를 고려하여 융합을 수행

02. Previous Work

3. Joint Cross-modality Image Registration Fusion

- **Unsupervised Misaligned Infrared and Visible Image Fusion via Cross-Modality Image Generation and Registration, 2022**



03

Methods

03. Methods

1. Overview

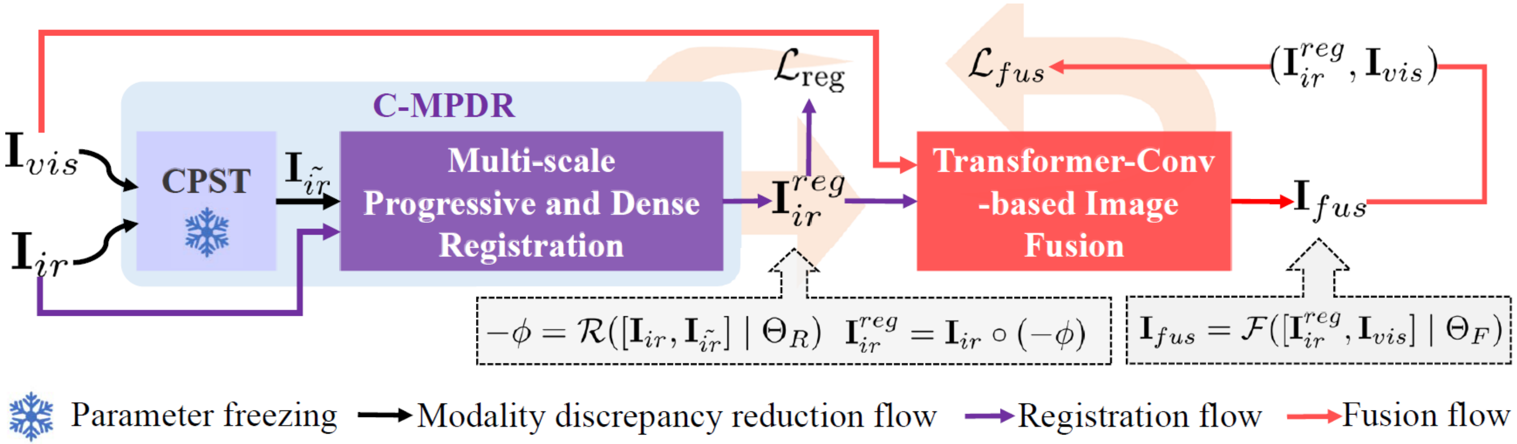


Fig. 2. Workflow of the proposed IMF for misaligned infrared and visible image fusion, which comprises two main portions. The first portion is a cross-modality multi-scale progressive and dense registration (C-MPDR) network, wherein the frozen CPST in UMF [19] mitigates cross-modality discrepancies and generates a pseudo-infrared image $\mathbf{I}_{\tilde{ir}}$. The MPDR subnetwork registers $\mathbf{I}_{ir}$ with $\mathbf{I}_{\tilde{ir}}$, producing a registered infrared image $\mathbf{I}_{ir}^{reg}$. The second portion is a Transformer-Conv-based fusion (TCF) subnetwork, which fuses $\mathbf{I}_{ir}^{reg}$ and $\mathbf{I}_{vis}$, generating the fused image $\mathbf{I}_{fus}$.

03. Methods

2. M-MDRFRegistration

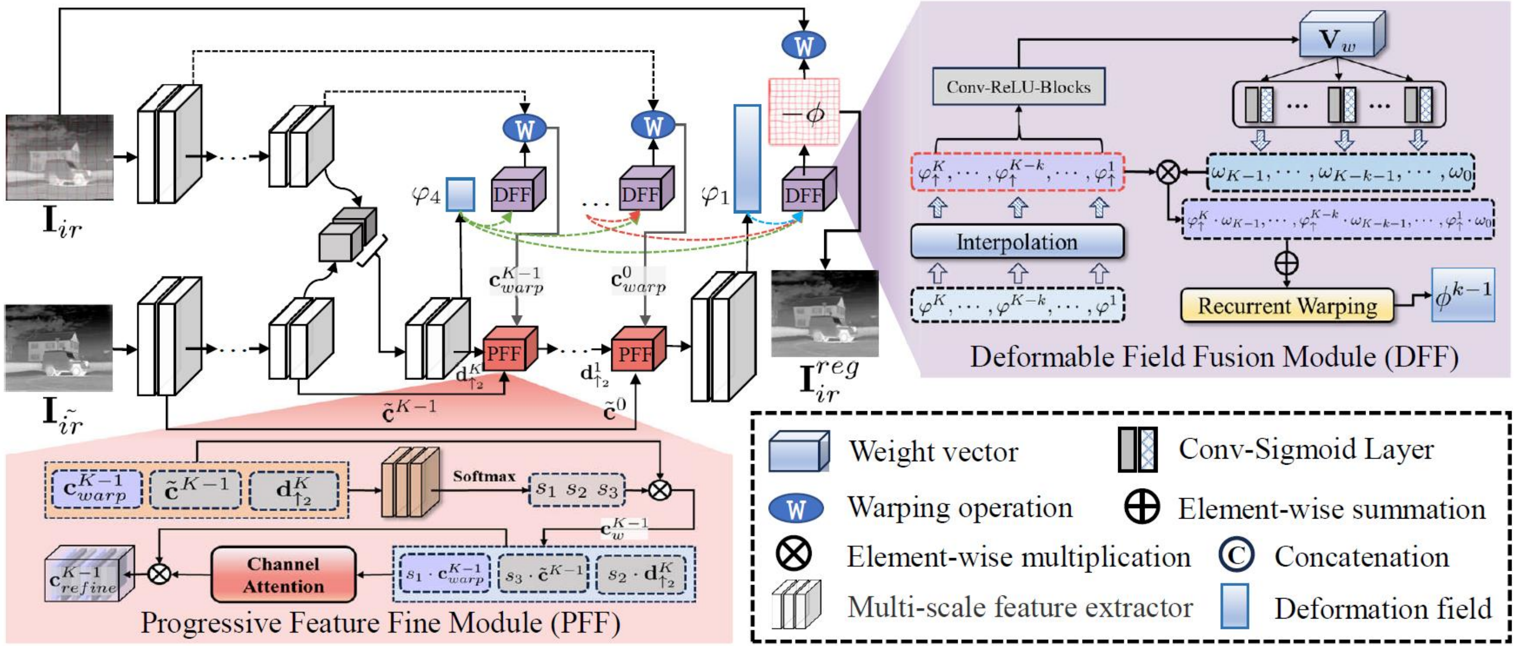
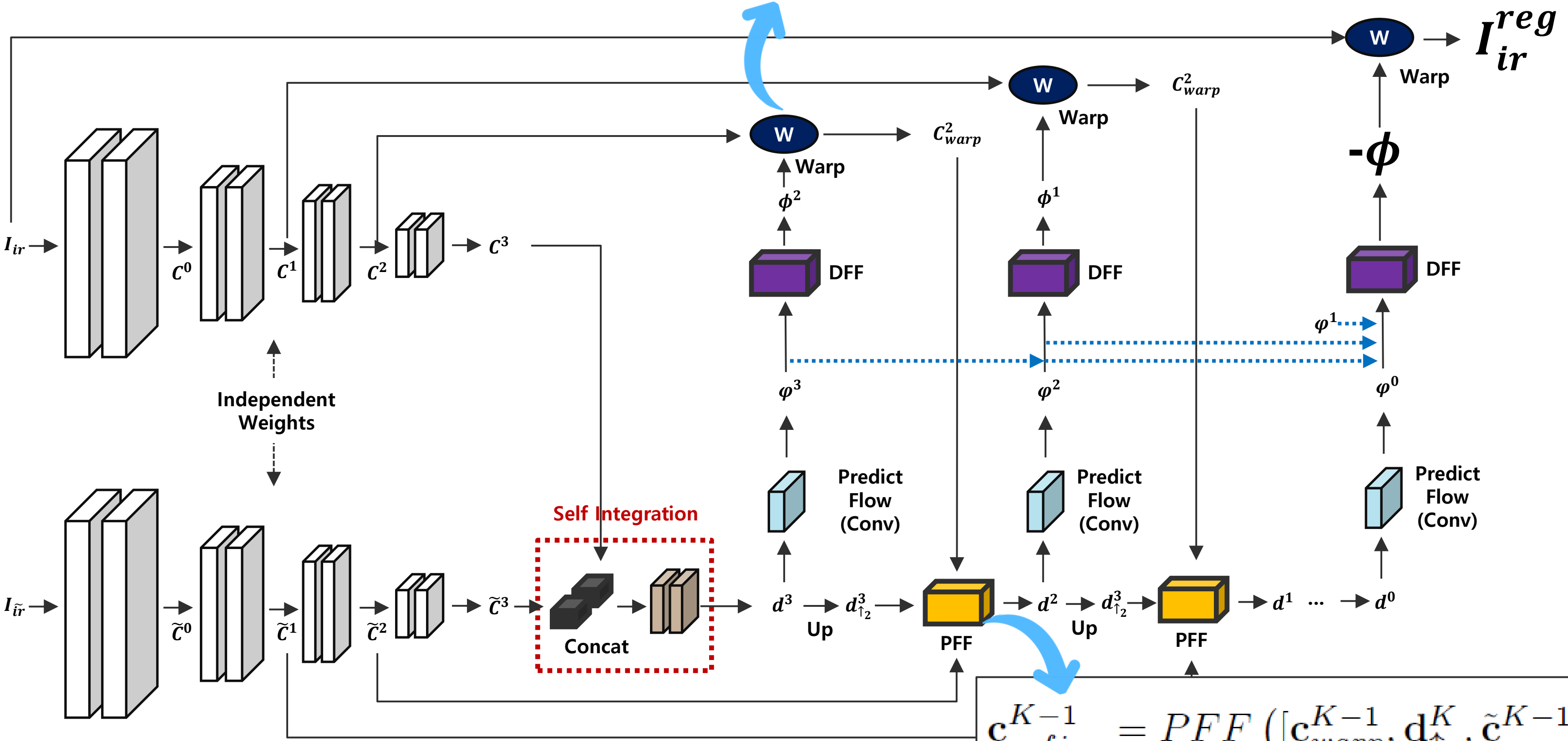


Fig. 3. Schematic of the proposed multi-scale progressive and dense registration (MPDR) subnetwork. Here, two independent feature extractors are utilized to extract hierarchical features from distorted-infrared and pseudo-infrared images, respectively. Subsequently, two core components, namely the deformable field fusion module (DFF) and the progressive feature fine module (PFF), collaborate to predict multi-scale deformation subfields in a dense propagation manner.

03. Methods

2. M-MDRF Registration

$$\mathbf{c}_{warp}^{K-1}(\mathbf{p}) = \mathbf{c}^{K-1}\left(\mathbf{p} + \phi^{K-1}(\mathbf{p})\right)$$

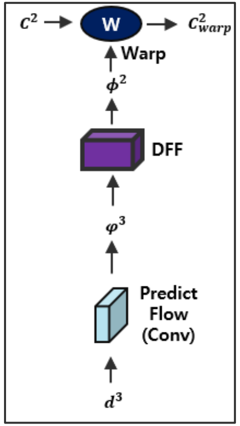


$$\mathbf{c}_{refine}^{K-1} = PFF\left(\left[\mathbf{c}_{warp}^{K-1}, \mathbf{d}_{\uparrow_2}^{K}, \tilde{\mathbf{c}}^{K-1}\right]\right)$$

03. Methods

2. M-MDRF$^{\text{Registration}}$: DFF Module

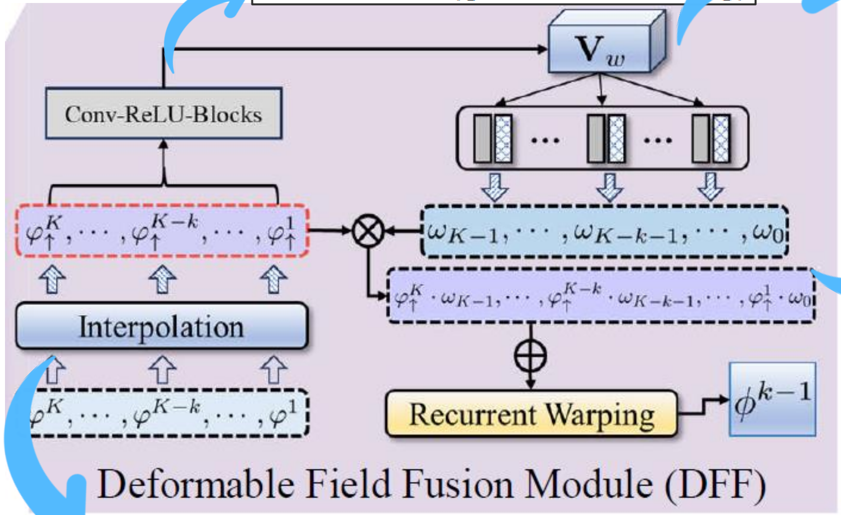
- **Coarse Registration**

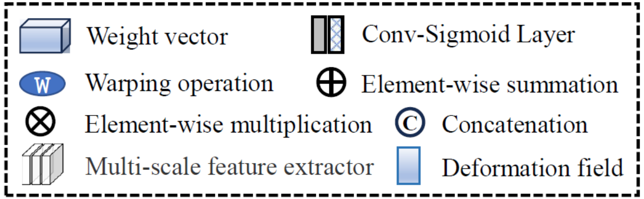


$$\mathbf{V}_w = \mathcal{M}_{\mathrm{CRB}}\left(\left[\varphi_\uparrow^K, \varphi_\uparrow^{K-1}, \cdots, \varphi_\uparrow^k\right]\right)$$

$$\omega_t = \mathcal{M}_{\mathrm{CSB}}[t]\left(\mathbf{V}_w\right)$$

$$\phi^{k-1} = \mathcal{V}\left(\sum_{i=k}^{K}\sum_{t=0}^{K-k}\omega_t \otimes \varphi_\uparrow^i\right)$$





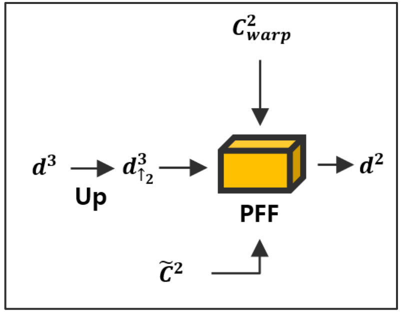
$$\varphi_\uparrow^i = \mathcal{P}_2^{i-k}\left(\varphi^i\right)$$

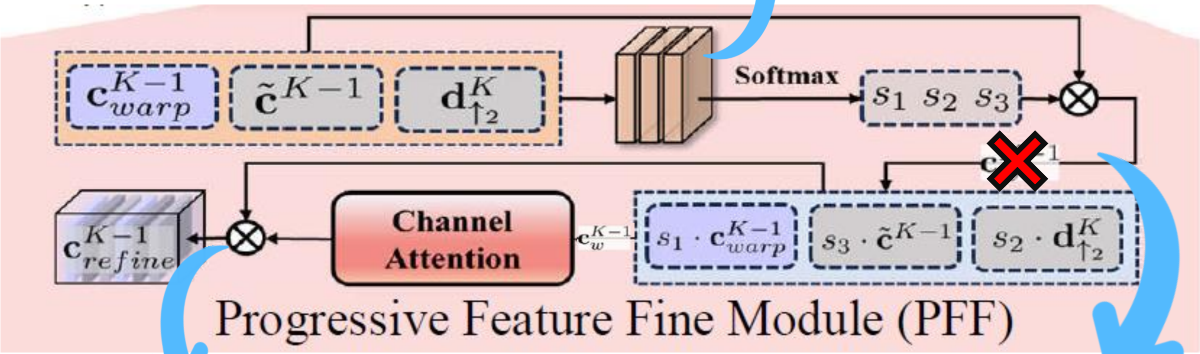
03. Methods

2. M-MDRF$^{\text{Registration}}$: PFF Module

- **Fine Registration**

$$s_1, s_2, s_3 = \mathtt{Softmax}\left(\mathcal{M}_{\mathrm{CRB}}(\mathbf{c}^{K-1}_{mix})\right)$$





$$\mathbf{c}^{K-1}_{refine} = \mathbf{c}^{K-1}_w \otimes \left(\mathcal{A}_{\mathtt{channel}}\left(\mathbf{c}^{K-1}_w\right)\right)$$

$$\mathbf{c}^{K-1}_w = \left[s_1 \cdot \mathbf{c}^{K-1}_{warp}, s_2 \cdot \mathbf{d}^{K}_{\uparrow_2}, s_3 \cdot \tilde{\mathbf{c}}^{K-1}\right]$$

03. Methods

2. M-MDRF Registration : Loss Function

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{sim} + \lambda_{sm}\mathcal{L}_{smooth}$$

$$\mathcal{L}_{sim}^{bi} = \|\psi_j(\mathbf{I}_{ir}^{reg}) - \psi_j(\mathbf{I}_{\tilde{ir}})\|_1 + \lambda_{rev}\|\psi_j(\phi \circ \mathbf{I}_{\tilde{ir}}) - \psi_j(\mathbf{I}_{ir})\|_1$$

$$\mathcal{L}_{\text{smooth}} = \|\nabla\phi\|_1$$

```python
def forward(self, y, y_f, tgt, src, flow, flow1, flow2):

    hyper_ncc = 1
    hyper_grad = 10
    hyper_feat = 1
    # TODO: similarity loss
    ncc_1 = torch.nn.functional.l1_loss(tgt, y)
    ncc_2 = torch.nn.functional.l1_loss(src, y_f)
```

```python
for (ir, it), (ir_path, it_path) in tqdm_loader:
    ir = ir.cuda()
    it = it.cuda()

    ir_affine, affine_disp = affine(ir)
    ir_elastic, elastic_disp = elastic(ir_affine)

    disp = affine_disp + elastic_disp
    ir_warp = ir_elastic

    ir_warp.detach_()
    disp.detach_()

    ir_pred, f_warp, flow, int_flow1, int_flow2, disp_pre = net(it, ir_warp)
    loss1, loss2, grad_loss, loss = _warp_Dense_loss_unsupervised(criterion, ir_pred, f_warp, it, ir_warp, flow, int_flow1, int_flow2)
```

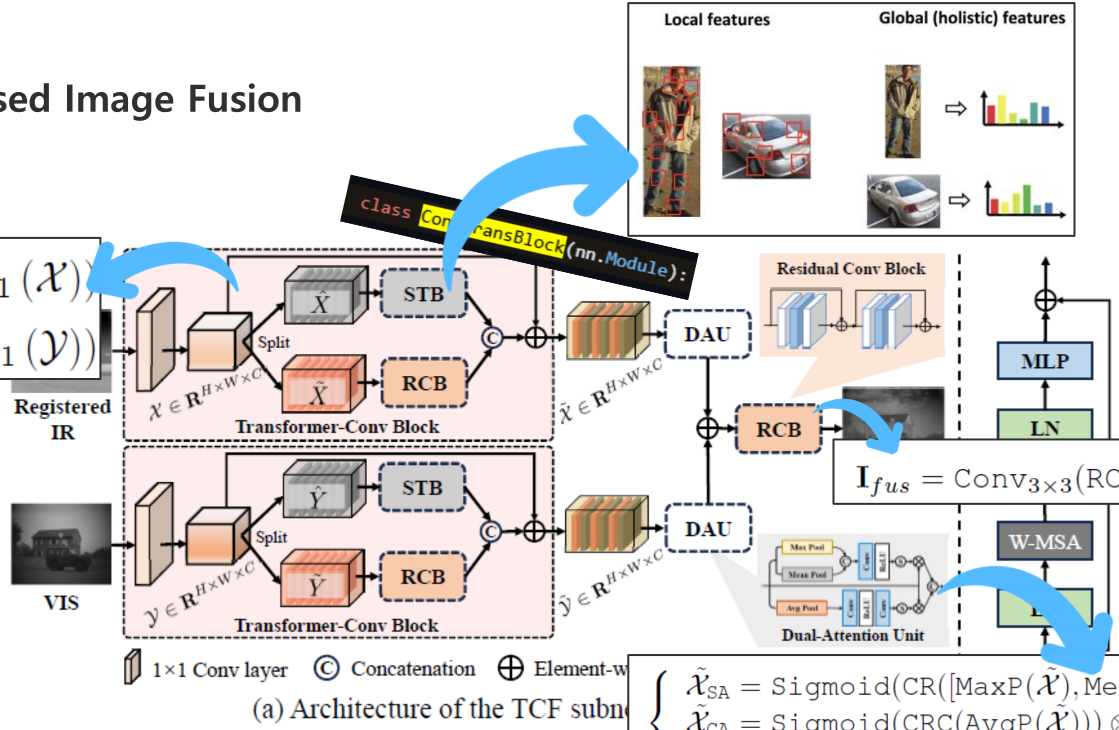
```python
# warping moving image
phi_4      = self.DFF_4(predict_flows)
final_flow = phi_4
warped_mov, disp_pre = self.spatial_transform_im(mov_img, final_flow)
warped_fix, _ = self.spatial_transform_im(fix_img, (-final_flow))

return warped_mov, warped_fix, final_flow, phi_3, phi_2, disp_pre
```

03. Methods

3. Transformer-Conv-based Image Fusion

$$\hat{\mathrm{X}}, \tilde{\mathrm{X}} = \mathtt{Split}\left(\mathtt{Conv}_{1\times1}\left(\mathcal{X}\right)\right)$$
$$\hat{\mathrm{Y}}, \tilde{\mathrm{Y}} = \mathtt{Split}\left(\mathtt{Conv}_{1\times1}\left(\mathcal{Y}\right)\right)$$



$$\mathbf{I}_{fus} = \mathtt{Conv}_{3\times3}(\mathtt{RConv}(\tilde{\mathcal{X}}_{\mathrm{EN}} + \tilde{\mathcal{Y}}_{\mathrm{EN}}))$$

$$\begin{cases} \tilde{\mathcal{X}}_{\mathrm{SA}} = \mathtt{Sigmoid}(\mathtt{CR}([\mathtt{MaxP}(\tilde{\mathcal{X}}), \mathtt{MeanP}(\tilde{\mathcal{X}})])) \otimes \tilde{\mathcal{X}}, \\ \tilde{\mathcal{X}}_{\mathrm{CA}} = \mathtt{Sigmoid}(\mathtt{CRC}(\mathtt{AvgP}(\tilde{\mathcal{X}}))) \otimes \tilde{\mathcal{X}}, \\ \tilde{\mathcal{X}}_{\mathrm{EN}} = \mathtt{CR}([\tilde{\mathcal{X}}_{\mathrm{SA}}, \tilde{\mathcal{X}}_{\mathrm{CA}}]), \end{cases}$$

Fig. 4. (a) The architecture of the Transformer-Conv-based image fusion subnetwork (TCFN). (b) Swin Transformer block. Noted that W-MSA refers to the multi-head self-attention mechanism with shifted windowing configuration.

03. Methods

3. Transformer-Conv-based Image Fusion : Loss Function

$$\mathcal{L}_{fus} = \lambda_{ssim}\mathcal{L}^{ms}_{ssim} + \lambda_{JG}\mathcal{L}_{JGrad} + \lambda_{svs}\mathcal{L}_{svs}$$

$$\mathcal{L}^{ms}_{ssim} = (1 - SSIM(\mathbf{I}_{fus}, \mathbf{I}^{reg}_{ir})) + (1 - SSIM(\mathbf{I}_{fus}, \mathbf{I}_{vis}))$$

$$\mathcal{L}_{JGrad} = \left\| max\left(\nabla\mathbf{I}^{reg}_{ir}, \nabla\mathbf{I}_{vis}\right), \nabla\mathbf{I}_{fus}\right\|_1$$

$$\omega_{ir} = S_{f_{ir}} / \left(S_{f_{ir}} - S_{f_{vis}}\right), \omega_{vis} = 1 - \omega_{ir},$$
$$\mathcal{L}_{svs} = \left\| \left(\omega_{ir} \otimes \mathbf{I}^{reg}_{ir} + \omega_{vis} \otimes \mathbf{I}_{vis}\right), I_{fus}\right\|_1,$$

```
def forward(self, im_fus, im_ir, im_vi, map_ir, map_vi, step):
    # ms_ssim_loss = (1 - self.ms_ssim(im_fus, im_ir)) + (1 - self.ms_ssim(im_fus, im_vi))
    ms_ssim_loss = (1 - self.ms_ssim(im_fus, (map_ir * im_ir + map_vi * im_vi)))

    # ssim_att_loss = self.ssim_att_loss(step, im_fus, (map_ir * im_ir + map_vi * im_vi))
    l1_loss       = self.l1_loss(im_fus, (map_ir * im_ir + map_vi * im_vi))
    grad_loss     = self.grad_loss(im_fus, im_ir, im_vi)
    fuse_loss     = self.alpha * ms_ssim_loss + self.beta * l1_loss + self.theta * grad_loss
    return fuse_loss
```

```
def forward(self, step, tgt, src):

    sigma = self.alpha * step + self.beta
    gauss = lambda x: torch.exp(-((x + 1) / sigma) ** 2) * self.maxVal
    ssim_map = self.ssimLoss(tgt, src, reduction='none').detach()
    gauss_map = gauss(ssim_map).detach()
    new_src, new_tgt = src * gauss_map, tgt * gauss_map

    loss = self.L1loss(new_src, new_tgt)

    return loss
```

04

Experiments

04. Experiments

1. Datasets

- **Synthethic misaligned IR/NIR-VIS Images**
  - IR-VIS Paired : TNO, RoadScene, M^3FD
  - NIR-VIS Paried : MSIFT
  - Random Transformation

    - Rotation(θ), Translation(t_x, t_y), Scaling(c_x, c_y), Shear(s_x, s_y)

$$\begin{bmatrix} c_x\cos(\theta) + s_x c_x \sin(\theta) & -c_x \sin(\theta) + s_x c_x \cos(\theta) & t_x \\ s_y c_y \cos(\theta) + c_y \sin(\theta) & -s_y c_y \sin(\theta) + c_y \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

TNO Dataset

RoadScene Dataset

M3FD Dataset

MSIFT Dataset

04. Experiments

1. Datasets

- **Real-world misaligned IR-VIS images**
  - FLIR Dataset

    - 10,228 IR-VIS image pairs from short Videos

    - 477 IR-VIS image pairs

FLIR Dataset

04. Experiments

2. Evaluation Metrics

- **Registration**

  - MSE Mean Square Error : 두 이미지 간 Pixel Value의 오차를 측정 $\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \tilde{y}_i)^2$

  - MI Mutual Information : Information theory 기반 이미지 내 Random Variables의 Independence를 측정

$$I(X;Y) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log_2\left(\frac{P(x,y)}{P(x)P(y)}\right)$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y),$$

  - NCC Normalized Cross Correlation : Matching Windows 간 Similarity를 측정 $NCC = \frac{1}{IJ}\sum_{j=1}^{J}\sum_{i=1}^{I}\frac{(A_{(i,j)}-a)(B_{(i,j)}-b)}{\sigma_A\sigma_B}$

- **Fusion**

  - CC Cross Correlation
  - MI Mutual Information
  - VIF Visual Information Fidelity : Mutual Information을 기반으로 정보량을 비교해 이미지품질 측정

$$\mathrm{VIF} = \frac{\sum_{j\in\mathrm{subbands}} I(\vec{C}^{N,j};\vec{F}^{N,j}|s^{N,j})}{\sum_{j\in\mathrm{subbands}} I(\vec{C}^{N,j};\vec{E}^{N,j}|s^{N,j})}$$

  - SSIM Structural Similarity Index : 구조적 유사도를 측정
  - Q_{abf} : Source Image에서 Fused Image로 Edge Information이 얼마나 전달됐는지 측정

04. Experiments

3. Results

- **Registration**

TABLE II

QUANTITATIVE EVALUATION OF CROSS-MODALITY IMAGE REGISTRATION ON THE TNO, ROADSCENE, M^3FD [2], AND MSIFT [39] DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED USING RED, BLUE AND GREEN IN THE ORDER. "†" DENOTES THAT THE CPST [19] IS TREATED AS A PLUGIN TO SEVERAL STATE-OF-THE-ART MONO-MODALITY OPTICAL FLOW MODELS TO IMPLEMENT CROSS-MODALITY FLOW ESTIMATION AND IMAGE REGISTRATION.

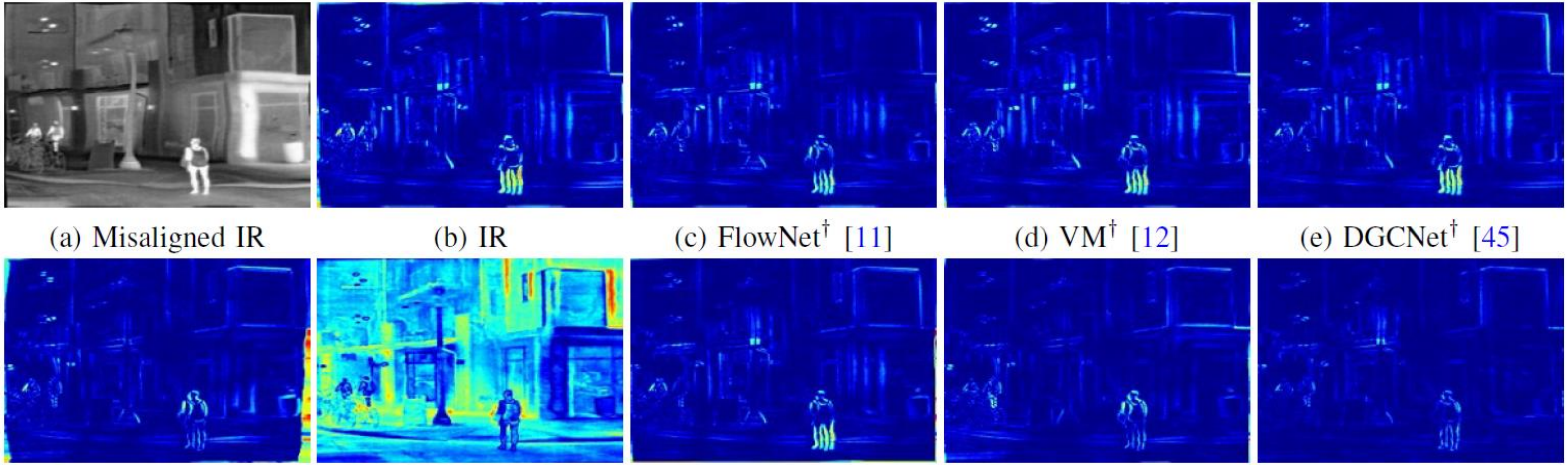
Methods	Public.	TNO			RoadScene			M^3FD			MSIFT		
		MSE ↓	NCC ↑	MI ↑	MSE ↓	NCC ↑	MI ↑	MSE ↓	NCC ↑	MI ↑	MSE ↓	NCC ↑	MI ↑
Misaligned	None	0.0067	0.876	1.558	0.0104	0.894	1.602	0.0054	0.915	2.054	0.0080	0.840	1.408
FlowNet [11]	CVPR'17	0.0292	0.636	1.095	0.0090	0.910	1.549	0.0357	0.636	1.638	**0.0014**	0.968	1.894
VoxelMorph [12]	CVPR'18	0.0069	0.880	1.545	0.0083	0.914	1.589	0.0051	0.922	2.009	0.0018	0.959	1.776
DGC-Net [45]	WACV'19	**0.0047**	**0.928**	1.606	0.0060	0.939	1.671	0.0032	0.950	2.116	0.0036	0.923	1.580
GLU-Net [46]	CVPR'20	0.0189	0.719	1.189	0.0127	0.867	1.467	0.0081	0.864	1.798	0.0053	0.894	1.473
FlowNet† [11]	CVPR'17	0.0073	0.893	1.465	**0.0051**	**0.949**	**1.744**	**0.0031**	**0.951**	**2.145**	**0.0014**	**0.969**	**1.899**
VoxelMorph† [12]	CVPR'18	0.0050	0.919	1.573	0.0058	0.941	1.689	0.0033	0.947	2.103	0.0018	0.960	1.787
DGC-Net† [45]	WACV'19	**0.0045**	**0.931**	**1.623**	0.0057	0.939	1.703	**0.0031**	**0.951**	2.128	0.0035	0.926	1.593
GLU-Net† [46]	CVPR'20	0.0197	0.736	1.257	0.0112	0.887	1.624	0.0085	0.861	1.792	0.0051	0.897	1.481
Nemar [33]	CVPR'20	0.1401	0.309	0.438	0.0814	0.846	0.986	0.1988	0.033	0.582	0.0848	0.704	0.689
CrossRAFT [47]	CVPR'22	0.0079	0.858	1.602	0.0088	0.910	**1.744**	0.0050	0.917	2.135	0.0059	0.888	1.787
SuperFusion [16]	CVPR'22	0.0070	0.886	1.507	0.0044	0.953	1.820	0.0034	0.942	2.076	0.0021	0.953	1.821
CGRP [19]	IJCAI'22	**0.0047**	0.926	**1.648**	**0.0036**	**0.963**	**1.833**	**0.0026**	**0.957**	**2.223**	**0.0013**	**0.970**	**1.897**
C-MPDR	None	**0.0028**	**0.957**	**1.804**	**0.0024**	**0.976**	**1.993**	**0.0018**	**0.970**	**2.243**	**0.0009**	**0.980**	**2.028**

04. Experiments

3. Results

- **Registration**

  - Synthetic Image



(a) Misaligned IR (b) IR (c) FlowNet† [11] (d) VM† [12] (e) DGCNet† [45]

(f) GLU-Net† [46] (g) Nemar [17] (h) CrossRAFT [47] (i) SuperFusion [16] (j) C-MPDR (Ours)

Fig. 5. Error visualization of the registration results on the **RoadScene** dataset. "†" denotes that the CPSTN [19] is utilized as the basic model to mitigate cross-modality discrepancy.

04. Experiments

3. Results

- **Registration**

  - Real World

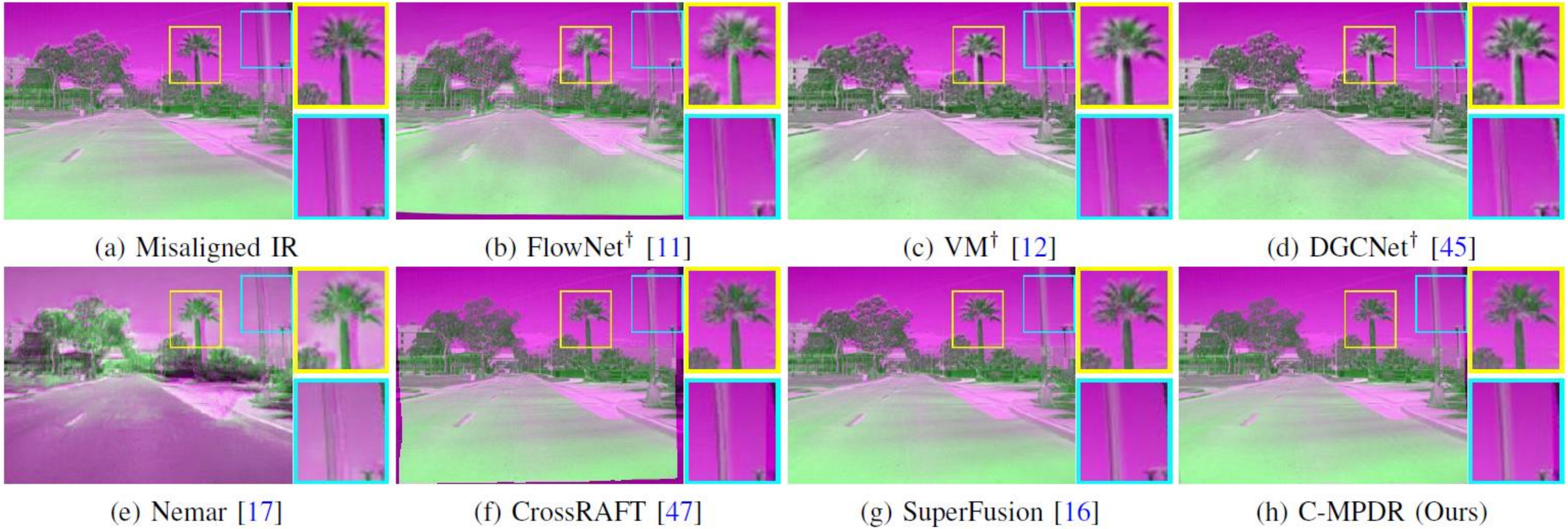


Fig. 6. Error visualization of the registration results on the **real-world FLIR** dataset. Noted that the CPSTN [19] is utilized as the basic model to mitigate cross-modality discrepancy.

04. Experiments

3. Results

- **Fusion**

TABLE III

THE QUANTITATIVE EVALUATION OF THE IVIF RESULTS WITH STATE-OF-THE-ART FUSION METHODS ON THREE DATASETS USING **FLOWNET** [11] ALGORITHM AS THE BASIC REGISTRATION MODEL. THE TOP THREE RESULTS ARE HIGHLIGHTED USING RED, BLUE AND GREEN IN THE ORDER.

Dataset	Metric	FGAN [4]	DIDFuse [5]	PMGI [6]	DDcGAN [7]	RFN [8]	MFEIF [9]	U2F [10]	TarDAL [2]	CDDFuse [1]	Ours
RoadScene	CC ↑	0.502	0.573	0.530	0.569	**0.587**	**0.588**	0.558	0.544	0.583	**0.595**
	MI ↑	1.886	1.942	1.965	1.941	1.951	2.009	1.681	**2.498**	**2.410**	**2.408**
	VIF ↑	0.493	0.445	0.408	0.538	0.563	**0.619**	0.430	**0.710**	0.609	**0.879**
	SSIM ↑	0.255	0.301	0.266	0.314	0.319	0.342	0.297	**0.435**	**0.468**	**0.499**
	Q_{abf} ↑	0.237	0.278	0.245	0.323	0.282	0.288	0.260	**0.360**	**0.424**	**0.427**
TNO	CC ↑	0.315	0.380	0.363	0.366	0.392	0.381	0.378	**0.440**	**0.430**	**0.481**
	MI ↑	1.368	1.762	1.489	1.434	1.628	1.743	1.329	**2.168**	**2.024**	**2.163**
	VIF ↑	0.588	0.545	0.518	0.510	0.671	**0.701**	0.535	**0.715**	0.567	**1.016**
	SSIM ↑	0.236	0.266	0.277	0.311	0.300	0.297	0.299	**0.437**	**0.338**	**0.473**
	Q_{abf} ↑	0.168	0.288	0.241	0.302	0.279	0.281	0.275	**0.389**	**0.331**	**0.427**
M^3FD	CC ↑	0.395	0.448	0.415	0.465	0.450	0.463	0.445	**0.482**	**0.500**	**0.530**
	MI ↑	1.816	2.162	2.043	1.925	2.080	2.210	1.889	**2.450**	**2.796**	**2.460**
	VIF ↑	0.596	0.555	0.519	0.451	0.479	**0.778**	0.486	0.643	**0.652**	**0.930**
	SSIM ↑	0.330	0.289	0.319	0.329	0.277	0.361	0.314	**0.395**	**0.442**	**0.483**
	Q_{abf} ↑	0.295	0.344	0.332	**0.386**	0.292	0.369	0.349	0.280	**0.450**	**0.417**

04. Experiments

3. Results

- **Registration**

  - Synthetic Image

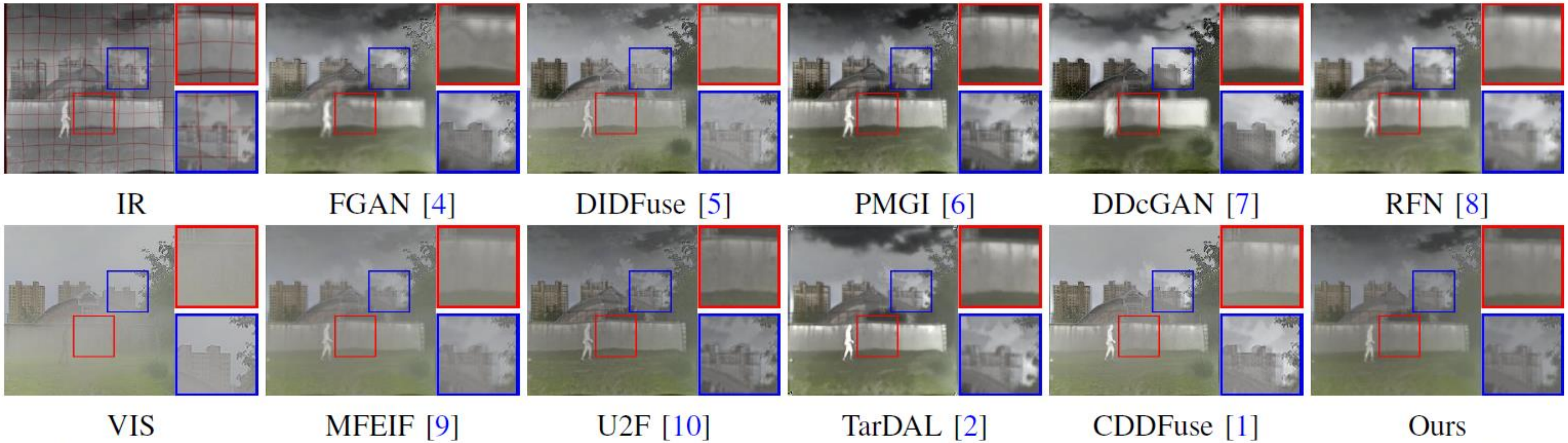


Fig. 9. Visualization of fusion results from the state-of-the-art infrared and visible image methods on the M^3FD dataset when using the **FlowNet** [11] as the basic registration model. Best viewed on screen.

04. Experiments

3. Results

- **Registration**
  - Real World

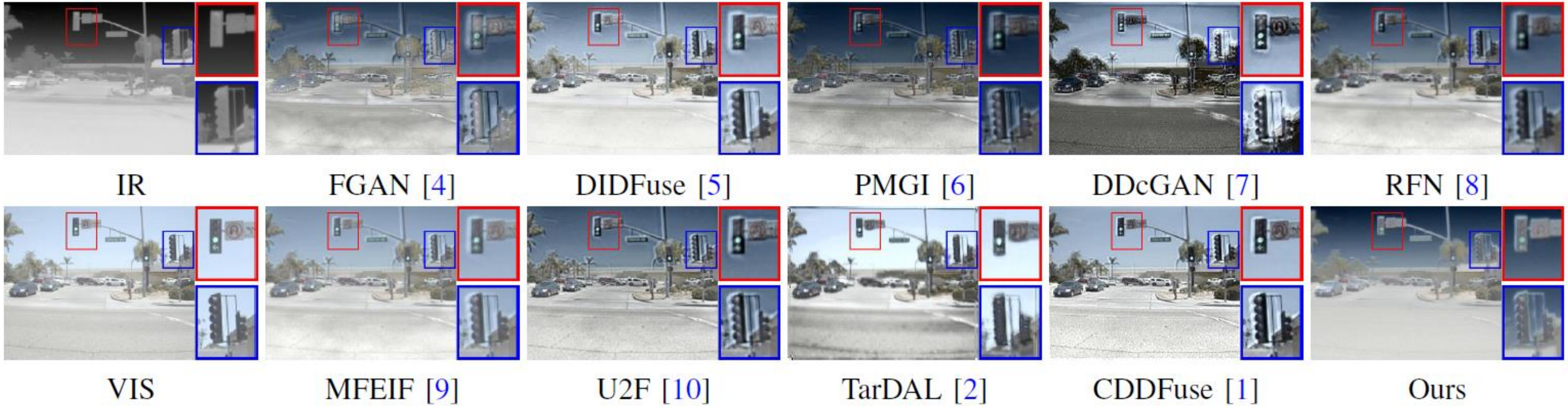


Fig. 10. Visualization of fusion results from the state-of-the-art infrared and visible image methods on the real-world misaligned *FLIR* dataset when using the **FlowNet** [11] as the basic registration model. Best viewed on screen.

04. Experiments

5. Ablation

- **C-MPDR**

TABLE IV
QUANTITATIVE ABLATION STUDIES OF THE C-MPDR NETWORK ON TNO, ROADSCENE, AND M^3FD DATASETS.

Dataset	**C-MPDR**	**TCF**	*Metrics* CC↑	SSIM↑	VIF↑
TNO	✗	✓	0.438	0.369	0.732
	✓	✓	**0.481** +0.044	**0.473** +0.103	**1.016** +0.374
Road	✗	✓	0.563	0.367	0.543
	✓	✓	**0.595** +0.032	**0.499** +0.132	**0.879** +0.336
M^3FD	✗	✓	0.493	0.348	0.663
	✓	✓	**0.530** +0.037	**0.483** +0.135	**0.930** +0.276

TABLE V
ABLATION STUDIES OF THE DFF AND PFF MODULES ON THE ROADSCENE DATASET.

No.	**DFF**	**PFF**	MSE↓	NCC↑	MI↑
1	Only Interpolation	Concatenation	0.0043	0.956	1.804
2	Only Interpolation	✓	0.0042	0.958	1.817
3	✓	Concatenation	0.0041	0.958	1.818
4	✓	✓	**0.0024**	**0.967**	**1.993**

04. Experiments

5. Ablation

- **TCF**

TABLE VI
ABLATION STUDIES OF THE TCB AND DAU MODULES ON THE ROADSCENE DATASET.

No.	TCB	DAU	CC↑	SSIM↑	VIF↑	MI↑	Q_{abf}↑
1	DenseBlock	✗	0.592	0.343	0.600	1.950	0.233
2	✓	✗	0.593	0.352	0.605	1.968	0.258
3	DenseBlock	✓	**0.596**	0.350	0.606	1.961	0.248
4	✓	✓	0.595	**0.499**	**0.879**	**2.408**	**0.427**

05

Conclusion

05. Conclusion

Contribution

- Multi Domain$^{IR\text{-}VIS}$ Registration 문제를 효과적으로 해결하는 방법을 제시
- Corse-to-fine 2단계를 End-to-End로 효과적으로 Registration 수행
- 효과적인 Registration 방법을 Fusion Network에 적용하여 고품질의 Fusion Image를 생성함

Limitation

- Target Task에 적용하여 Fusion Image의 효과를 제시하면 본 논문 방법의 진보성을 부각 가능
- 방법에 대한 설명과 실험에 대한 내용이 더 보강되면 이해하기 쉬울 것

Discussion

- 코드와 논문이 어디부분까지 일치해야 하는가 ?
- Domain Transfer를 통해 Image를 Translation하기 때문에 Domain 변환 네트워크의 성능에 영향 클 것 예상
  - 제안하는 방법이 불가능한 도메인이 존재할 것
  - 엄밀히 보면, Image Transfer – Registration의 두 단계. 1개의 Network로 확장할 수 있는 방법?

Q & A

Thank you for your attention